

J KARTHIK

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CSA0911

JAVA

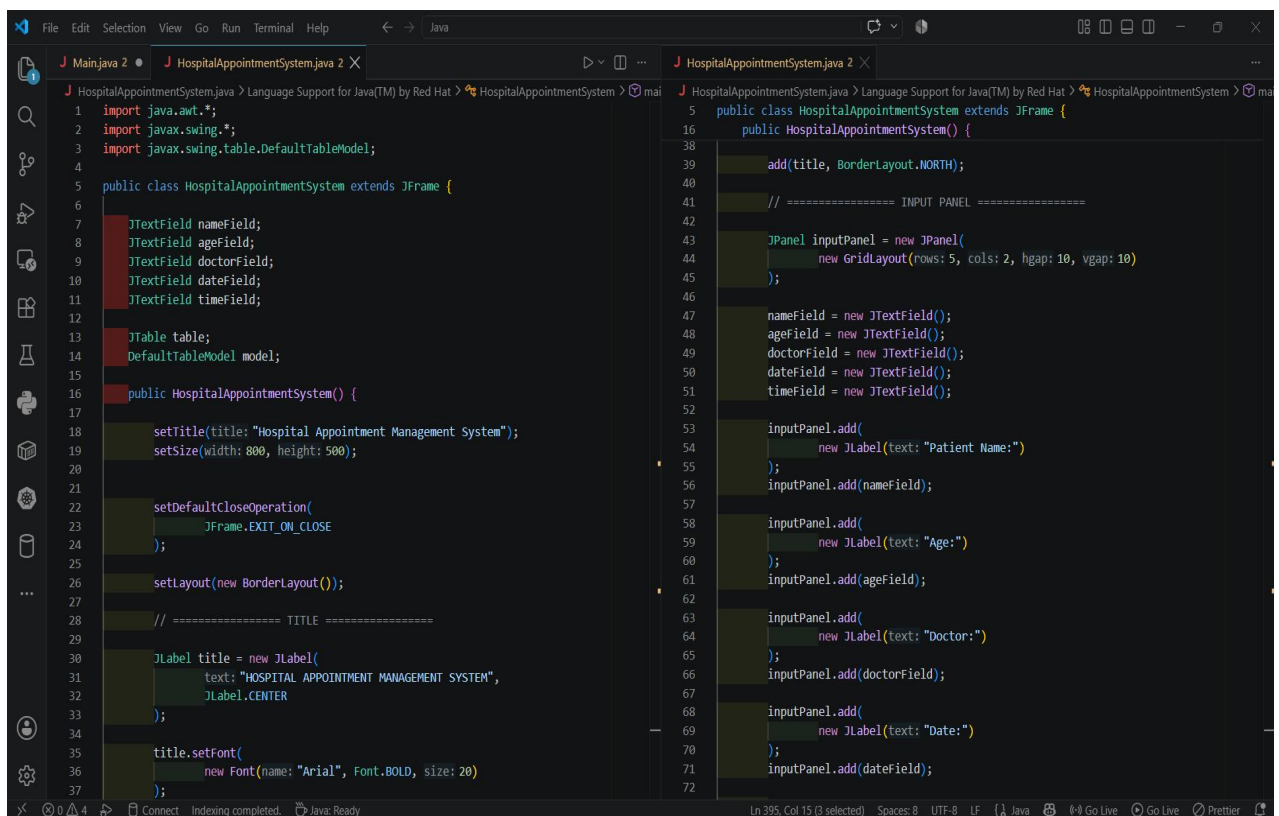
CO4 AT1 Q2

Q2 Hospital Appointment Management System

50 marks

Design a Swing-based GUI application for a hospital to manage patient appointments, including booking, updating, and canceling schedules. Use AWT/Swing controls such as buttons, text fields, tables, and menus to facilitate user interaction. Implement event listeners to handle actions like appointment confirmation and deletion. Ensure the system can handle dynamic patient data efficiently. Evaluate different layout managers (FlowLayout, GridLayout, BorderLayout) for organizing complex UI components. Analyze how the design supports scalability when the number of patients increases. Propose improvements for better usability and real-time updates.

CODE:



```
1 import java.awt.*;
2 import javax.swing.*;
3 import javax.swing.table.DefaultTableModel;
4
5 public class HospitalAppointmentSystem extends JFrame {
6
7     JTextField namefield;
8     JTextField agefield;
9     JTextField doctorfield;
10    JTextField datefield;
11    JTextField timefield;
12
13    JTable table;
14    DefaultTableModel model;
15
16    public HospitalAppointmentSystem() {
17
18        setTitle(title: "Hospital Appointment Management System");
19        setSize(width: 800, height: 500);
20
21
22        setDefaultCloseOperation(
23            JFrame.EXIT_ON_CLOSE
24        );
25
26        setLayout(new BorderLayout());
27
28        // ===== TITLE =====
29
30        JLabel title = new JLabel(
31            text: "HOSPITAL APPOINTMENT MANAGEMENT SYSTEM",
32            JLabel.CENTER
33        );
34
35        title.setFont(
36            new Font(name: "Arial", Font.BOLD, size: 20)
37        );
38
39        add(title, BorderLayout.NORTH);
40
41        // ===== INPUT PANEL =====
42
43        JPanel inputPanel = new JPanel(
44            new GridLayout(rows: 5, cols: 2, hgap: 10, vgap: 10)
45        );
46
47        namefield = new JTextField();
48        agefield = new JTextField();
49        doctorfield = new JTextField();
50        datefield = new JTextField();
51        timefield = new JTextField();
52
53        inputPanel.add(
54            new JLabel(text: "Patient Name:")
55        );
56        inputPanel.add(namefield);
57
58        inputPanel.add(
59            new JLabel(text: "Age:")
60        );
61        inputPanel.add(agefield);
62
63        inputPanel.add(
64            new JLabel(text: "Doctor:")
65        );
66        inputPanel.add(doctorfield);
67
68        inputPanel.add(
69            new JLabel(text: "Date:")
70        );
71        inputPanel.add(datefield);
72
73    }
```

```
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J Main.java 2 J HospitalAppointmentSystem.java 2 X
J HospitalAppointmentSystem.java > Language Support for Java(TM) by Red Hat > HospitalAppointmentSystem > mai
5 public class HospitalAppointmentSystem extends JFrame {
16 public HospitalAppointmentSystem() {
73 inputPanel.add(
74     new JLabel(text: "Time:");
75 );
76 inputPanel.add(timeField);
77
78 add(inputPanel, BorderLayout.WEST);
79
80 // ===== TABLE =====
81
82 String[] columns = {
83     "Patient Name",
84     "Age",
85     "Doctor",
86     "Date",
87     "Time"
88 };
89
90 model = new DefaultTableModel(
91     columns,
92     rowCount: 0
93 );
94
95 table = new JTable(model);
96
97 add(
98     new JScrollPane(table),
99     BorderLayout.CENTER
100 );
101
102 // ===== BUTTONS =====
103
104 JPanel buttonPanel = new JPanel();
105
106 JButton book =
107     new JButton(text: "Book");
108
109 JButton update =
110     new JButton(text: "Update");
111
112 JButton cancel =
113     new JButton(text: "Cancel");
114
115 JButton clear =
116     new JButton(text: "Clear");
117
118 buttonPanel.add(book);
119 buttonPanel.add(update);
120 buttonPanel.add(cancel);
121 buttonPanel.add(clear);
122
123 add(
124     buttonPanel,
125     BorderLayout.SOUTH
126 );
127
128 // ===== BOOK =====
129
130 book.addActionListener(e -> bookAppointment());
131
132 // ===== UPDATE =====
133
134 update.addActionListener(
135     e -> updateAppointment()
136 );
137
138 // ===== CANCEL =====
139
140 cancel.addActionListener(
141     e -> cancelAppointment()
142 );
143
Ln 395, Col 15 (3 selected) Spaces: 8 UTF-8 LF Java Go Live Go Live Prettier
```

```
File Edit Selection View Go Run Terminal Help Java
J Main.java 2 J HospitalAppointmentSystem.java 5 J HospitalAppointmentSystem.java 5
italAppointmentSystem.java > Language Support for Java(TM) by Red Hat > HospitalAppointmentSystem > main(String[])
5 public class HospitalAppointmentSystem extends JFrame {
16 public HospitalAppointmentSystem() {
146 clear.addActionListener(
147     e -> clearFields()
148 );
149
150 // ===== TABLE SELECTION =====
151
152 table.getSelectionModel()
153     .addListSelectionListener(e -> {
154
155         int row =
156             table.getSelectedRow();
157
158         if (row >= 0) {
159
160             nameField.setText(
161                 model.getValueAt(
162                     row, column: 0
163                 ).toString()
164             );
165
166             ageField.setText(
167                 model.getValueAt(
168                     row, column: 1
169                 ).toString()
170             );
171
172             doctorField.setText(
173                 model.getValueAt(
174                     row, column: 2
175                 ).toString()
176             );
177
178             dateField.setText(
179                 model.getValueAt(
180                     row, column: 3
181
182             );
183
184             timeField.setText(
185                 model.getValueAt(
186                     row, column: 4
187                 ).toString()
188             );
189
190         });
191
192 setLocationRelativeTo(c; null);
193 setVisible(b: true);
194
195 // ===== BOOK APPOINTMENT =====
196
197 void bookAppointment() {
198
199     try {
200
201         String name =
202             nameField.getText().trim();
203
204         String age =
205             ageField.getText().trim();
206
207         String doctor =
208             doctorField.getText().trim();
209
210         String date =
211             dateField.getText().trim();
212
213         String time =
214             timeField.getText().trim();
215
Ln 395, Col 17 Spaces: 8 UTF-8 LF Java Go Live Go Live Prettier
```

```
5 public class HospitalAppointmentSystem extends JFrame {
198 void bookAppointment() {
217     if (name.isEmpty())
218         || age.isEmpty()
219         || doctor.isEmpty()
220         || date.isEmpty()
221         || time.isEmpty()) {
222
223         throw new Exception(
224             message: "Please enter all details."
225         );
226     }
227
228     Integer.parseInt(age);
229
230     model.addRow(
231         new Object[]{
232             name,
233             age,
234             doctor,
235             date,
236             time
237         }
238     );
239
240     JOptionPane.showMessageDialog(
241         this,
242         message: "Appointment Booked Successfully!"
243     );
244
245     clearFields();
246 }
247 } catch (NumberFormatException e) {
248
249     JOptionPane.showMessageDialog(
250         this,
251         message: "Age must be a number."
252     );
253 }
254 } catch (Exception e) {
255
256     JOptionPane.showMessageDialog(
257         this,
258         e.getMessage()
259     );
260 }
261 }
262
263 // ===== UPDATE =====
264
265 void updateAppointment() {
266
267     int row =
268         table.getSelectedRow();
269
270     if (row == -1) {
271
272         JOptionPane.showMessageDialog(
273             this,
274             message: "Please select an appointment."
275         );
276     }
277
278     return;
279 }
280
281 try {
282     String name =
283         nameField.getText().trim();
284
285     String age =
286         ageField.getText().trim();
287
288     String doctor =
```

```
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italAppointmentSystem.java > Language Support for Java(TM) by Red Hat > HospitalAppointmentSystem > main(String[])
5 public class HospitalAppointmentSystem extends JFrame {
265 void updateAppointment() {
291     String date =
292         dateField.getText().trim();
293
294     String time =
295         timeField.getText().trim();
296
297     if (name.isEmpty()
298         || age.isEmpty()
299         || doctor.isEmpty()
300         || date.isEmpty()
301         || time.isEmpty()) {
302
303         throw new Exception(
304             message: "Please enter all details."
305         );
306     }
307
308     Integer.parseInt(age);
309
310     model.setValueAt(
311         name, row, column: 0
312     );
313
314     model.setValueAt(
315         age, row, column: 1
316     );
317
318     model.setValueAt(
319         doctor, row, column: 2
320     );
321
322     model.setValueAt(
323         date, row, column: 3
324     );
325
326     model.setValueAt(
327         time, row, column: 4
328     );
329
330     JOptionPane.showMessageDialog(
331         this,
332         message: "Appointment Updated Successfully!"
333     );
334
335     clearFields();
336
337 } catch (NumberFormatException e) {
338
339     JOptionPane.showMessageDialog(
340         this,
341         message: "Age must be a number."
342     );
343
344 } catch (Exception e) {
345
346     JOptionPane.showMessageDialog(
347         this,
348         e.getMessage()
349     );
350
351 }
352
353 // ===== CANCEL =====
354
355 void cancelAppointment() {
356
357     int row =
358         table.getSelectedRow();
359
360     if (row == -1) {
```

```
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J Main.java 2 J HospitalAppointmentSystem.java 5

J HospitalAppointmentSystem.java > Language Support for Java(TM) by Red Hat > HospitalAppointmentSystem > main(String[])
5 public class HospitalAppointmentSystem extends JFrame {
355 void cancelAppointment() {
362 JOptionPane.showMessageDialog(
363     this,
364     message: "Please select an appointment."
365 );
366
367 return;
368 }
369
370 model.removeRow(row);
371
372 JOptionPane.showMessageDialog(
373     this,
374     message: "Appointment Cancelled Successfully!"
375 );
376
377 clearFields();
378 }
379
380 // ===== CLEAR =====
381
382 void clearFields() {
383
384     nameField.setText(t: "");
385     ageField.setText(t: "");
386     doctorField.setText(t: "");
387     dateField.setText(t: "");
388     timeField.setText(t: "");
389
390     table.clearSelection();
391 }
392
393 // ===== MAIN =====
394
395 public static void main(String[] args) {
```

```
File Edit Selection View Go Run Terminal Help Java

J Main.java 2 J HospitalAppointmentSystem.java 5

J HospitalAppointmentSystem.java > Language Support for Java(TM) by Red Hat > HospitalAppointmentSystem > main(String[])
5 public class HospitalAppointmentSystem extends JFrame {
    Run | Debug | Run main | Debug main
395 public static void main(String[] args) {
396
397     SwingUtilities.invokeLater(() ->
398         new HospitalAppointmentSystem()
399     );
400 }
401 }
```

OUTPUT:

Hospital Appointment Management System

Patient Name:

Age:

Doctor:

Date:

Time:

Patient Name

Age

Doctor

Date

Time

Karthik

19

Sai

22-08-26

10 45

Book

Update

Cancel

Clear